Bezawit Tinsae Birhane

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Professional Summary

Mechanical Engineering student with a 4.0 GPA and strong experience in mechanical design, CAD modeling, and engineering simulation. Certified SOLIDWORKS (CSWA & CSWP) user with hands-on experience in mechanical system design, thermal analysis (ANSYS/COMSOL), and MATLAB/Simulink modeling. Published researcher with proven ability to translate engineering concepts into validated prototypes and analytical models. Passionate about aerospace systems, propulsion concepts, and advanced mechanical design, and eager to apply strong problem-solving and technical skills in innovative engineering environments.

EXPERIENCE

Research Assistant (PI: Dr. Yasir) | Higher Colleges of Technology | March 2025 – Present

- Developed an intelligent EV charging optimization model reducing peak grid load by 27%.
- Engineered a custom-controlled Solid Oxide Fuel Cell (SOFC) system improving biomass efficiency by 36.31%.
- Applied GIS tools for smart grid infrastructure optimization.
- Co-authored and presented two papers at IEEE PES ISGT Middle East 2025.

Engineering Projects
Team Lead | 3DCP Thermal Research (Advisor: Dr. Abraham Mansuri) | Higher Colleges of Technology | Jan 2025 – Nov 2025

- Developed a low-cost IR–Arduino system for non-contact R-value measurement in 3D printed concrete.
- Designed a dual-servo SolidWorks mechanism for automated temperature mapping.
- Validated accuracy with 0.61% relative error against benchmarks.
- Led a 4-member team to Wiley Heat Transfer Journal publication.

Capstone Project – Autonomous Inventory Management Robot Higher Colleges of Technology | 2026 – Present

- Designing an **autonomous robotic system** for pharmacy inventory monitoring.
- Integrating **sensing, navigation logic, and mechanical system design**.
- Developing system simulations and control models using **MATLAB/Simulink**.
- Applying engineering design principles to optimize system efficiency and reliability.

EDUCATION

Higher Colleges of Technology (HCT) | *UAE* **Bachelor of Applied Science in Mechanical Engineering** | *2023 – Present*

- **Academic Standing:** Cumulative GPA: 4.0/4.0 .
- **Scholarship:** Awarded Full International Scholarship after ranking among the top students in the 2022 Ethiopian National Exam.
- **Technical Design:** Proficient in SolidWorks and Mastercam for advanced mechanical

modelling and manufacturing.

- **Theoretical Foundation:** Strong expertise in Thermodynamics, Applied Dynamics, Mechanical Design, Mechanics of Materials, and Statistics.

Course Projects

- Designed a **complete conveyor belt system using SolidWorks**, including full assembly and component-level mechanical simulations for shafts, bearings, and fasteners.
- Performed **heat transfer analysis using ANSYS and COMSOL** to study thermal performance under different boundary conditions.
- Developed **dynamic control system models using MATLAB and Simulink**

SKILLS

- **Engineering & Design:** SolidWorks, Mastercam, 3D Printing (3DCP), Prototyping.
- **Analysis & Programming:** Ansys, COMSOL, Python, MATLAB, Arduino, Statistics.
- **Energy Systems:** Smart Grid Modelling, SOFC, Infrared Sensing, Thermodynamics.
- **Leadership & Collaboration:** Experienced Team Lead for multidisciplinary research; skilled in project management and cross-functional teamwork.
- **Public Speaking: Toastmasters Dubai** member; Fifth standing in **EEG** public speaking competitions.
- **Communication:** Expert Technical Writing (IEEE/Wiley published) and advanced **PowerPoint** for professional research dissemination.
- **Problem Solving:** Proven ability to validate complex systems with high precision (0.61% error margins) under strict deadlines.

Certificates

- **CSWA & CSWP**: Certified SOLIDWORKS Associate and Professional.
- **AI & Computing:** KAUST Artificial Intelligence Training Certificate.
- **Energy Systems:** Demand Response and the Smart Grid (Certificate).
- **Simulation:** Master COMSOL Multiphysics Certificate.
- **Management**: Project Management Agile Skills Certificate.

Professional Membership

- **Student Affiliate Member** – Institution of Mechanical Engineers (IMechE)
- **Student Member** – ASHRAE UAE Chapter
- **Student Member** – IEEE
- **Member** – Toastmasters Dubai Legends Club

Leadership Experience

- **Event Organization Leader**: Golden Key International Honour Society – HCT | 2024 – Present
    - Lead planning and execution of academic and professional development events on campus.
    - Coordinate cross-functional student teams to ensure successful event delivery.
- **Project Team Leader**: Higher Colleges of Technology | 2023 – Present
    - Led multiple academic engineering projects, managing design, simulation, and reporting phases.
    - Supervised team coordination, task delegation, and milestone achievement.
- **President – Volunteer Club:** University of Gondar Community School [2021 – 2022]
    - Led student volunteer initiatives and managed community service projects.